

Design and VLSI Implementation of a Hybrid Parallel Prefix Adder Based on Han-Carlson and Ladner-Fischer Architectures Using CMOS and Advanced GDI Techniques

A Comparative Analysis of CMOS, PTL, DPTL, GDI, MGDI, and FSGDI Logic Styles Through Transistor-Level and 32nm Physical Layout Implementation

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Abstract - The surging demand for high-speed and power frugal arithmetic circuits has driven research into optimized parallel prefix adder architectures and advanced logic design techniques. This paper presents the design and VLSI implementation of a Hybrid Han-Carlson-Ladner-Fischer Prefix Adder, integrating the sparse carry computation of the Han-Carlson architecture with the balanced carry distribution of the Ladner-Fischer prefix network. The proposed architecture is implemented and comparatively evaluated using both traditional CMOS and advanced non-conventional logic styles, namely Pass Transistor Logic (PTL), Dual Pass Transistor Logic (DPTL), Gate Diffusion Input (GDI), Modified Gate Diffusion Input (MGDI), and Full-Swing Gate Diffusion Input (FSGDI). We perform transistor-level implementation and switching analysis using DSCH. The physical layout is analyzed in Microwind using 32 nm CMOS technology, selected as a balanced technology node to achieve an effective compromise between silicon area, power dissipation, and implementation complexity. Comparative analysis demonstrates that the proposed hybrid architecture, when implemented using advanced GDI-based logic styles, achieves up to 60.17% lower power consumption, 24.00% lower propagation delay, and 35.00% smaller silicon area compared with the conventional CMOS implementation, optimized area utilization, and reduced propagation delay while maintaining reliable arithmetic functionality. The proposed design is therefore well suited for high-performance and low-power VLSI arithmetic systems.

Keywords - Hybrid Parallel Prefix Adder, Han-Carlson Adder, Ladner-Fischer Adder, CMOS Logic, PTL, DPTL, Gate Diffusion Input (GDI), Modified GDI (MGDI), Full-Swing GDI (FSGDI), DSCH, Microwind, 32 nm CMOS Technology, Low-Power VLSI, High-Speed Arithmetic, Parallel Prefix Architecture

I. INTRODUCTION

The rapid evolution of Very Large Scale Integration (VLSI) technology has transformed the design of digital integrated circuits by enabling higher transistor densities and improved computational capability. As electronic systems become increasingly sophisticated, arithmetic circuits are expected to deliver faster operation while maintaining low power

consumption and efficient utilization of silicon resources. Among the fundamental building blocks of digital hardware, binary adders occupy a critical position because they are extensively employed in arithmetic and logic units (ALUs), digital signal processors, microprocessors, embedded systems, communication hardware, cryptographic processors, and artificial intelligence accelerators. Since addition forms the basis of numerous arithmetic operations, the efficiency of the adder directly influences the overall performance of the complete system.

Continuous technology scaling has introduced several design constraints that challenge conventional arithmetic circuits. Increased leakage current, interconnection complexity, propagation delay, switching power, and reduced noise immunity have become major concerns in nanometer-scale implementations. Consequently, considerable research has been devoted to developing adder architectures that simultaneously improve computational speed, minimize energy dissipation, and reduce hardware overhead. Among the available alternatives, Parallel Prefix Adders (PPAs) have gained significant attention because they perform carry generation through parallel prefix operations, thereby shortening the critical path and offering superior performance compared with conventional ripple carry and carry look-ahead adders.

Several Parallel Prefix Adder topologies have been proposed, each emphasizing a different balance between logic depth, wiring complexity, fan-out, and hardware utilization. The Han-Carlson Adder employs a sparse prefix configuration that effectively limits interconnect congestion while maintaining fast carry computation. In contrast, the Ladner-Fischer Adder utilizes a balanced prefix tree capable of generating carries with comparatively lower logic depth and efficient signal distribution. Although both architectures demonstrate excellent computational characteristics, neither independently provides the optimum trade-off among speed, silicon area, routing complexity, and power consumption.

Combining the advantageous features of these two prefix structures provides an effective strategy for achieving a more balanced and resource-efficient arithmetic architecture.

Besides architectural refinement, the underlying logic implementation significantly affects circuit's performance. Conventional Complementary Metal-Oxide-Semiconductor (CMOS) logic is widely adopted because of its stable operation, excellent noise tolerance, and dependable output characteristics. However, the relatively high transistor count associated with CMOS implementations often increases area occupancy and dynamic power consumption. To address these limitations, several non-conventional logic families have been investigated. Pass Transistor Logic (PTL) minimizes transistor usage by transmitting logic signals through switching transistors, while Dual Pass Transistor Logic (DPTL) improves voltage transfer capability through complementary transistor operation. Gate Diffusion Input (GDI) technology further decreases transistor count by implementing multiple Boolean functions using compact circuit configurations. Modified Gate Diffusion Input (MGDI) enhances the practicality of GDI implementations by improving output reliability, whereas Full-Swing Gate Diffusion Input (FSGDI) restores complete voltage swing to improve signal integrity without introducing substantial hardware overhead. These alternative logic styles provide attractive opportunities for designing energy-efficient arithmetic circuits suitable for advanced VLSI applications.

This research proposes a Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder that combines the sparse carry computation capability of the Han-Carlson structure with the balanced prefix network of the Ladner-Fischer architecture. The objective is to achieve a practical compromise among propagation delay, power dissipation, transistor count, and layout area without sacrificing computational accuracy. To evaluate the influence of different circuit implementation methodologies, the proposed architecture is realized using CMOS, PTL, DPTL, GDI, MGDI, and FSGDI logic styles. Functional verification and transistor-level schematic design are performed using the DSCH environment, whereas physical layout generation and performance evaluation are carried out in Microwind utilizing 32 nm CMOS technology. The selected technology node offers an appropriate balance between fabrication complexity, integration density, switching performance, and power efficiency, making it well suited for comparative VLSI investigations.

Performance evaluation confirms that integrating advanced GDI-based logic techniques with the proposed hybrid prefix architecture produces noticeable improvements in power consumption, propagation delay, transistor utilization, and layout efficiency while maintaining accurate arithmetic functionality. These characteristics indicate that the proposed design is an effective solution for implementing high-speed and low-power arithmetic units intended for modern VLSI systems where performance, energy efficiency, and hardware optimization are equally important.

The remaining sections of this paper are organized as follows. Section II presents the literature review and discusses the fundamental concepts related to Parallel Prefix Adders and various logic implementation techniques. Section III describes the existing methodology, including conventional Parallel

Prefix Adder architectures and previously adopted logic styles such as CMOS, PTL, DPTL, GDI, and MGDI, along with their limitations in terms of power consumption, delay, and area optimization. Section IV presents the proposed methodology, which introduces the Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder architecture and its implementation using advanced FSGDI logic. The proposed design approach, transistor-level implementation, and physical layout realization using DSCH and Microwind with 32 nm technology are discussed in this section. Section V presents the simulation environment and design flow, including schematic design using DSCH and layout generation using Microwind. Section VI discusses the simulation results and comparative performance analysis, including functional verification, power consumption, propagation delay, silicon area, and overall performance evaluation. Section VII highlights the ease of use, design flexibility, and implementation aspects of the proposed architecture. Finally, Section VIII concludes the work by summarizing the major findings and presenting future scope for further research, followed by the references.

Parallel Prefix Adder Architecture

A Parallel Prefix Adder (PPA) performs binary addition by computing carry signals in parallel, thereby reducing propagation delay compared with conventional adder architectures. The carry computation process is divided into three fundamental stages: Pre-processing, Carry Generation (Prefix Computation), and Post-processing.

A. Pre-processing Stage

The pre-processing stage generates the propagate (P) and generate (G) signals for each input bit pair. These signals determine whether a carry is propagated or generated during addition.

The Boolean expressions are given by:

Generate Signal (G):

$$G_i = A_i \cdot B_i$$

Propagate Signal (P):

$$P_i = A_i \oplus B_i$$

B. Carry Generation (Prefix Computation)

The carry generation stage computes the carry signals in parallel using the prefix network. Adjacent propagate and generate pairs are recursively combined through black and gray prefix cells to obtain higher-order carries with reduced logic depth.

The prefix operation is represented as,

Group Generate:

$$G_{i:j} = G_i + (P_i \cdot G_j)$$

Group Propagate:

$$P_{i:j} = P_i \cdot P_j$$

or equivalently,

$$(G_i, P_i) \circ (G_j, P_j) = (G_i + P_i G_j, P_i P_j)$$

C. Post-processing Stage

The post-processing stage produces the final sum bits using the carry signals generated by the prefix network.

The Boolean equations are

Sum Equation:

$$S_i = P_i \oplus C_i$$

Final Carry:

$$C_{i+1} = G_i + (P_i \cdot C_i)$$

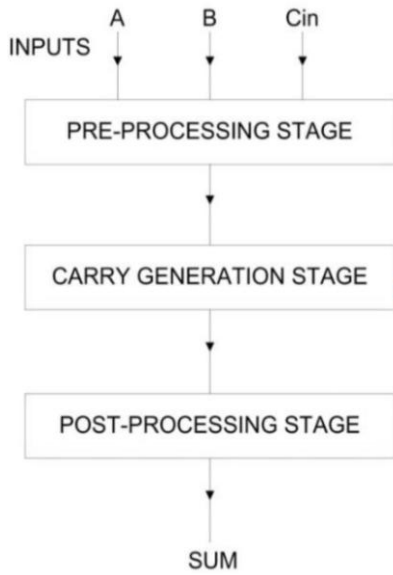


Fig.1: Design flow of Parallel Prefix Adder

II. LITERATURE REVIEW

[1] K. Srilatha and Avala Mallikarjuna Prasad (2025) investigated the implementation of a Ladner-Fischer Parallel Prefix Adder using PTL, DPTL, and GDI logic styles. The transistor-level circuits were developed in DSCH 3, and the corresponding physical layouts were generated using Microwind 3.1. Their comparative analysis indicated that GDI minimized silicon area, whereas DPTL exhibited lower power consumption. However, the study was restricted to the Ladner-Fischer architecture and did not explore CMOS, MGDI, FSGDI, or hybrid Parallel Prefix Adder designs.

[2] Rekha P., Smitha Gayathri D., Yasha Jyothi M. Shirur, Shamanth K. S., Kumar P. Gowda, and Bindu S. (2025) presented a 32-bit Arithmetic Logic Unit implemented using Full-Swing Gate Diffusion Input (FS-GDI) logic. The proposed design was compared with CMOS, GDI, and MGDI implementations, demonstrating improvements in transistor count, power consumption, and Power Delay Product (PDP). Nevertheless, the investigation focused exclusively on ALU implementation and did not consider Parallel Prefix Adder architectures.

[3] Nitin Singh (2025) proposed a low-power 4-bit Arithmetic Logic Unit based on the Modified Gate Diffusion Input (MGDI) technique. The transistor-level schematic was implemented using DSCH 3.5, and the physical layout was generated in Microwind employing 65 nm CMOS technology. Experimental results showed noticeable reductions in switching delay, transistor count, and power dissipation compared with conventional CMOS logic. However, the work was limited to ALU design and did not investigate Parallel Prefix Adders.

[4] Sehan Amandu Gamage, Krishnan Subramaniam, and Ahmad Anwar Zainuddin (2022) conducted a comparative analysis of CMOS-based Full Adders using DSCH and Microwind across different CMOS technology nodes. Their transistor-level implementation and layout evaluation demonstrated that technology scaling improves propagation delay and silicon utilization. Although the study validated the effectiveness of CMOS logic, advanced non-conventional logic techniques such as PTL, GDI, MGDI, and FSGDI were not considered.

[5] Muhibul Haque Bhuyan, Md. Mahfuz Ahmed, and Shafiul Alam Robin (2021) designed and analyzed a 4-bit CMOS Full Adder using DSCH and Microwind at 90 nm, 65 nm, and 45 nm technology nodes. The comparative study revealed that technology scaling significantly reduces power consumption, propagation delay, and layout area while improving circuit performance. However, the proposed work was confined to CMOS implementation and did not investigate Parallel Prefix Adder architectures or alternative logic styles.

[6] Harsh Yadav, Amit Kumar Goyal, and Ajay Kumar (2020) presented a Gate Diffusion Input (GDI) based Full Adder and compared its performance with conventional CMOS implementation. Post-layout simulation results demonstrated lower transistor count, reduced propagation delay, decreased power consumption, and improved energy efficiency. However, the proposed design was limited to a single-bit full adder and was not extended to complex arithmetic structures such as Parallel Prefix Adders.

[7] Majid Amini-Valashani and Sattar Mirzakuchaki (2020) introduced Modified Gate Diffusion Input (MGDI) based Full Adder cells for energy-efficient VLSI applications. The transistor-level implementation achieved full-swing operation with lower power consumption and improved Power Delay Product compared with conventional CMOS circuits. Despite these advantages, the work focused only on basic full adder

design and did not address hybrid Parallel Prefix Adder architectures or comprehensive comparisons among multiple logic styles.

From the reviewed literature, it is evident that previous research has primarily concentrated on CMOS, GDI, MGDI, and FS-GDI implementations of Full Adders and Arithmetic Logic Units using DSCH and Microwind. Only limited attention has been given to transistor-level implementation of Parallel Prefix Adders using advanced non-conventional logic techniques, and no comprehensive study has combined the Han-Carlson and Ladner-Fischer architectures with CMOS, PTL, DPTL, GDI, MGDI, and FSGDI logic styles under a unified 32 nm DSCH and Microwind design environment. Therefore, the proposed Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder addresses this research gap by providing a detailed transistor-level and layout-level comparative analysis focusing on power consumption, propagation delay, and silicon area optimization.

B. Implementation and Evaluation

The complete design flow is carried out using DSCH for transistor-level schematic development and functional verification, followed by physical layout realization in Microwind using 32 nm CMOS technology. These widely used VLSI design tools simplify circuit simulation, transistor-level validation, and layout generation, allowing efficient evaluation without requiring highly specialized design environments.

Furthermore, the hybrid architecture supports systematic performance comparison across multiple logic styles under identical technology conditions. This simplifies design exploration and assists researchers in identifying the most suitable implementation technique for low-power and high-speed arithmetic applications. Owing to its modular organization, compatibility with standard VLSI tools, and efficient implementation methodology, the proposed design can be readily adopted for educational purposes, research activities, and practical digital integrated circuit development.

III. EXISTING METHODOLOGY

The implementation of arithmetic circuits has undergone continuous improvement with the advancement of Very Large Scale Integration (VLSI) technology. Among various arithmetic components, Parallel Prefix Adders (PPAs) have become a preferred solution for high-speed computation because they perform carry generation in parallel, thereby reducing the critical path delay compared with conventional ripple carry and carry-look-ahead adders. Existing implementations predominantly utilize individual prefix architectures, including the Han-Carlson Adder and the Ladner-Fischer Adder, together with conventional and non-conventional logic design techniques. Although these methodologies provide considerable improvements in computational performance, each implementation exhibits specific limitations related to power consumption, propagation delay, transistor utilization, layout complexity, and routing efficiency.

1. Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder

A. CMOS Logic Implementation

Complementary Metal-Oxide-Semiconductor (CMOS) logic has remained the standard implementation methodology for digital integrated circuits because of its excellent reliability, full voltage swing, high noise immunity, and stable switching characteristics. In Parallel Prefix Adders, CMOS logic ensures accurate carry generation and dependable arithmetic operation under different operating conditions. The pull-up and pull-down transistor networks provide balanced output transitions, making CMOS highly suitable for robust digital circuit implementation.

Despite these advantages, CMOS implementation requires a comparatively larger number of transistors to realize complex prefix structures such as Han-Carlson and Ladner-Fischer Adders. The increased transistor count contributes to larger silicon area, higher dynamic switching power, higher leakage current, and greater routing complexity. As technology scales toward nanometer dimensions, these limitations become more significant, reducing the suitability of conventional CMOS for ultra-low-power VLSI systems.

B. Pass Transistor Logic (PTL)

Pass Transistor Logic (PTL) was introduced to reduce transistor utilization by transmitting logic signals directly through switching transistors instead of employing complementary pull-up and pull-down networks. By minimizing transistor count, PTL decreases silicon area, parasitic capacitance, and dynamic power dissipation while improving implementation compactness. These characteristics make PTL attractive for arithmetic circuits where hardware optimization is an important design objective.

However, PTL suffers from threshold voltage degradation during signal transmission, resulting in incomplete logic levels and reduced noise margins. Cascaded PTL circuits often require additional restoration circuitry to recover degraded output voltages, thereby increasing design complexity. Consequently, although PTL improves hardware efficiency, maintaining reliable signal integrity in high-speed Parallel Prefix Adders remains challenging.

C. Dual Pass Transistor Logic (DPTL)

Dual Pass Transistor Logic (DPTL) extends the PTL methodology by employing complementary NMOS and PMOS transistors for signal propagation. Using both types of transistors for signal propagation improves voltage transfer characteristics, restores logic levels more effectively, and reduces threshold voltage loss encountered in conventional PTL circuits. DPTL therefore provides improved switching performance and enhanced reliability while maintaining relatively low transistor utilization.

Although DPTL offers better voltage characteristics than PTL, the requirement for complementary transistor pairs increases routing complexity and implementation overhead. As the arithmetic circuit size increases, additional routing resources and transistor connections can limit improvements in silicon area and overall design efficiency.

D. Gate Diffusion Input (GDI)

Gate Diffusion Input (GDI) is an efficient non-conventional logic design technique capable of implementing numerous Boolean functions using a considerably smaller number of transistors than conventional CMOS circuits. The compact circuit structure reduces switching capacitance, resulting in lower dynamic power consumption, smaller silicon area, and improved propagation characteristics. Owing to these advantages, GDI has become an attractive solution for low-power arithmetic circuit implementation.

Despite its benefits, conventional GDI implementation may experience reduced output voltage swing and fabrication constraints under standard CMOS manufacturing processes. Signal degradation can affect circuit reliability, particularly in large arithmetic architectures consisting of multiple cascaded logic stages. Therefore, additional optimization techniques are often required to improve practical implementation.

E. Modified Gate Diffusion Input (MGDI)

Modified Gate Diffusion Input (MGDI) was developed to overcome several limitations associated with conventional GDI implementation. By modifying transistor connections and improving voltage transfer characteristics, MGDI enhances signal integrity while preserving the low transistor count of GDI circuits. Consequently, MGDI achieves improved power efficiency, lower propagation delay, reduced layout area, and better implementation flexibility compared with CMOS, PTL, DPTL, and conventional GDI.

Several transistor-level implementations using MGDI have demonstrated improvements in arithmetic circuits such as full adders and arithmetic logic units. However, most existing studies concentrate on small-scale arithmetic blocks and do not investigate the application of MGDI to complex Parallel Prefix Adder architectures. Furthermore, comprehensive comparative analysis between CMOS, PTL, DPTL, GDI, and MGDI under identical simulation conditions using DSCH and Microwind has received limited attention.

F. Limitations of Existing Methodology

The existing implementation methodologies provide valuable improvements in arithmetic circuit performance; however, several challenges remain unresolved. Conventional CMOS implementation requires a high transistor count, resulting in larger silicon area, increased power consumption, and higher propagation delay. PTL reduces hardware complexity but suffers from threshold voltage degradation and incomplete voltage transfer. DPTL improves voltage restoration but introduces additional routing complexity and implementation

overhead. Although GDI and MGDI significantly decrease transistor utilization and improve power efficiency, their applications are predominantly restricted to full adders, multiplexers, and arithmetic logic units rather than large-scale Parallel Prefix Adder architectures.

Moreover, existing studies generally investigate individual logic styles independently without providing a comprehensive transistor-level comparison under identical technology conditions. A unified implementation of a Hybrid Han-Carlson–Ladner-Fischer Parallel Prefix Adder employing CMOS, PTL, DPTL, GDI, MGDI, and advanced Full-Swing Gate Diffusion Input (FSGDI) using DSCH and Microwind at the 32 nm CMOS technology node has not been comprehensively reported. These limitations provide the motivation for the proposed methodology, which aims to achieve reduced power consumption, lower propagation delay, optimized transistor utilization, and compact layout while maintaining reliable arithmetic functionality.

2. Logic Style Implementation (CMOS, PTL, DPTL, GDI, and MGDI)

The implementation methodology of Parallel Prefix Adders is significantly influenced by the selected logic design technique, as it directly affects transistor count, power consumption, propagation delay, and silicon area. Existing arithmetic circuits have primarily been realized using conventional CMOS logic and several non-conventional logic families, including Pass Transistor Logic (PTL), Dual Pass Transistor Logic (DPTL), Gate Diffusion Input (GDI), and Modified Gate Diffusion Input (MGDI). Each implementation technique offers distinct advantages and limitations with respect to circuit performance and hardware efficiency.

Conventional CMOS logic provides excellent voltage swing, high noise immunity, and reliable switching characteristics, making it the most widely adopted design methodology for digital integrated circuits. However, the larger transistor count required for complex arithmetic structures increases silicon area, switching power, and propagation delay.

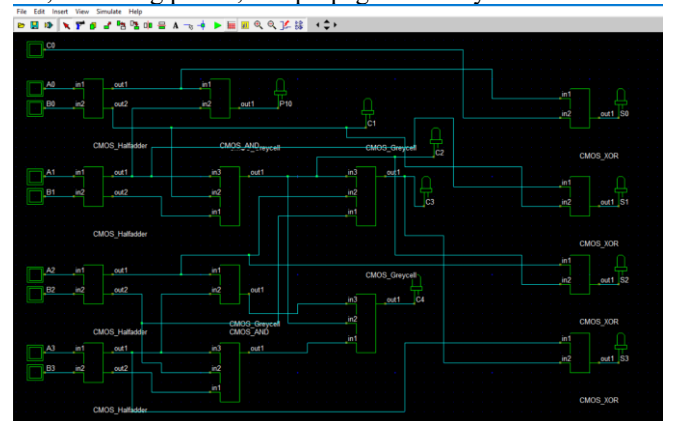


Fig.2: Implementation of HPA using CMOS in DSCH

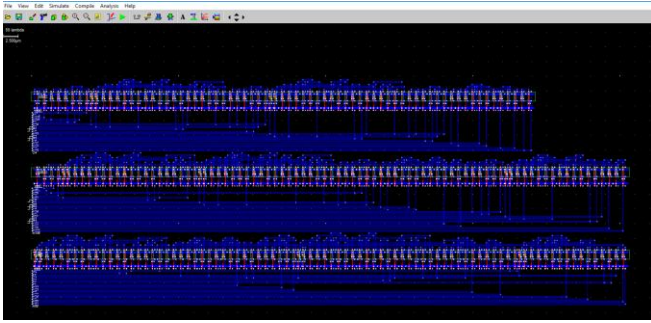


Fig.3: Layout of HPA using CMOS in Microwind

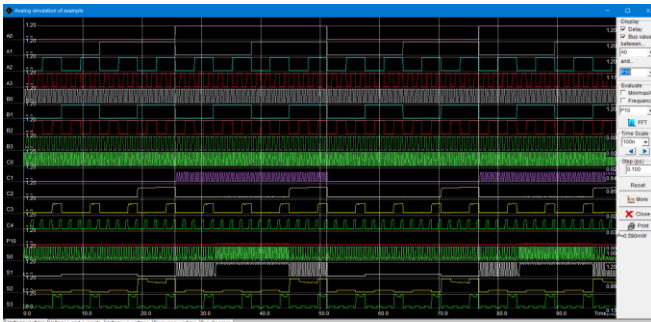


Fig.4: Power Analysis of HPA using CMOS in Microwind

Pass Transistor Logic (PTL) minimizes transistor utilization by transmitting logic signals directly through switching transistors, resulting in compact circuit implementation and reduced dynamic power consumption. Nevertheless, threshold voltage degradation and reduced output voltage levels may affect signal integrity in cascaded arithmetic circuits.

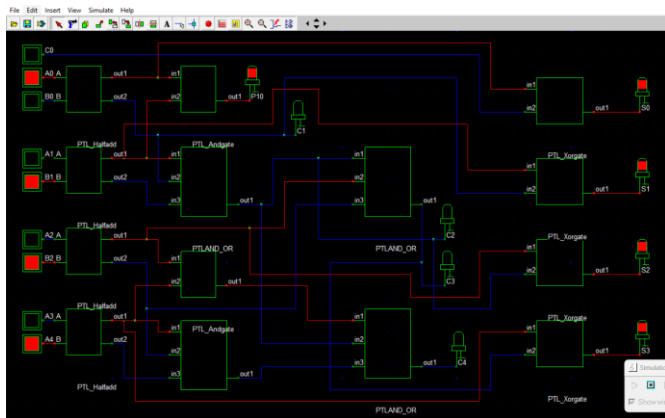


Fig.5: Implementation of HPA using PTL in DSCH

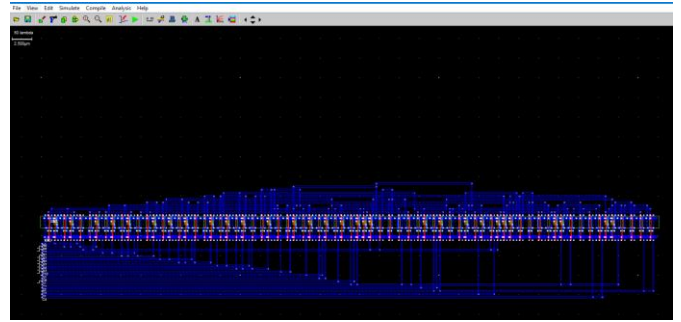


Fig.6: Layout of HPA using PTL in Microwind

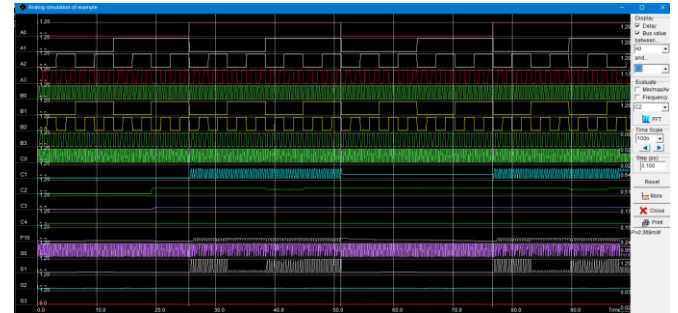


Fig.7: Power Analysis of HPA using PTL in Microwind

Dual Pass Transistor Logic (DPTL) overcomes several limitations of PTL by employing complementary NMOS and PMOS transistors for signal transmission. This technique improves voltage restoration and switching reliability while maintaining relatively low transistor utilization. However, additional transistor connections increase routing complexity and layout overhead.

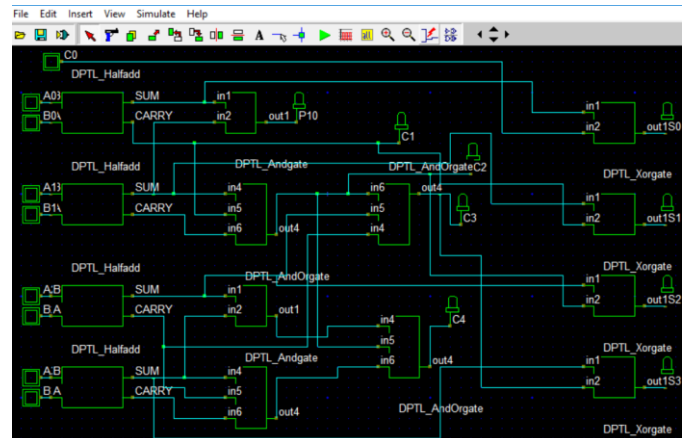


Fig.8: Implementation of HPA using DPTL in DSCH

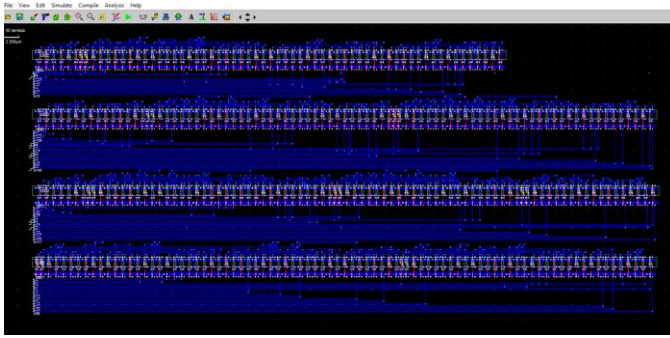


Fig.9: Layout of HPA using DPTL in Microwind

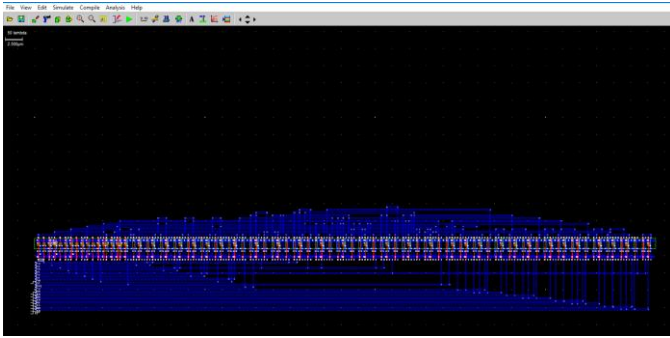


Fig.12: Layout of HPA using GDI in Microwind

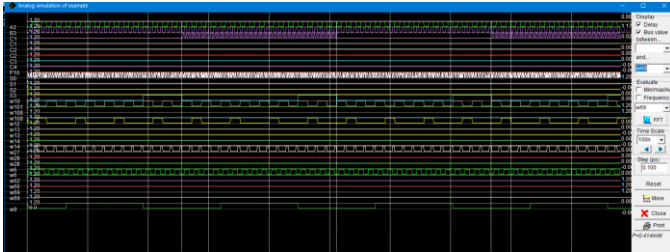


Fig.10: Power Analysis of HPA using DPTL in Microwind

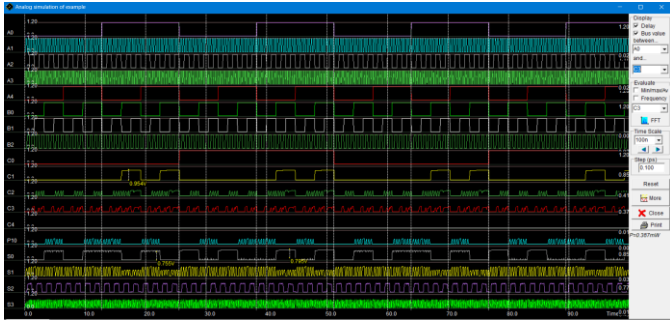


Fig.13: Power Analysis of HPA using GDI in Microwind

Gate Diffusion Input (GDI) significantly reduces transistor count by implementing multiple logic functions using compact transistor configurations. The reduction in switching capacitance contributes to lower power consumption, reduced propagation delay, and smaller silicon area. Despite these advantages, conventional GDI circuits may experience reduced output voltage swing under standard CMOS fabrication conditions.

Modified Gate Diffusion Input (MGDI) enhances the GDI methodology by improving output voltage characteristics and signal reliability without considerably increasing hardware complexity. MGDI achieves better power efficiency, lower propagation delay, and improved layout utilization compared with CMOS, PTL, DPTL, and conventional GDI. However, existing implementations have mainly focused on basic arithmetic circuits, while comprehensive transistor-level implementation of Hybrid Parallel Prefix Adders remains limited.

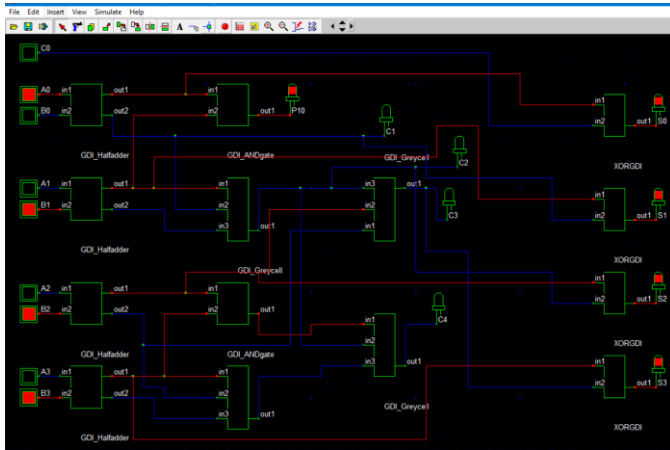


Fig.11: Implementation of HPA using GDI in DSCH

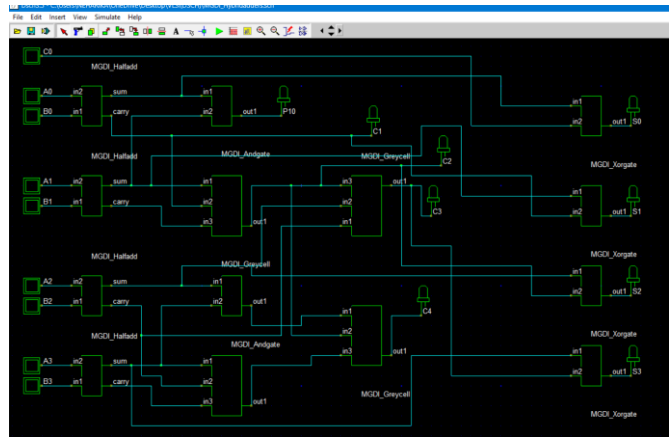


Fig.14: Implementation of HPA using MGDI in DSCH

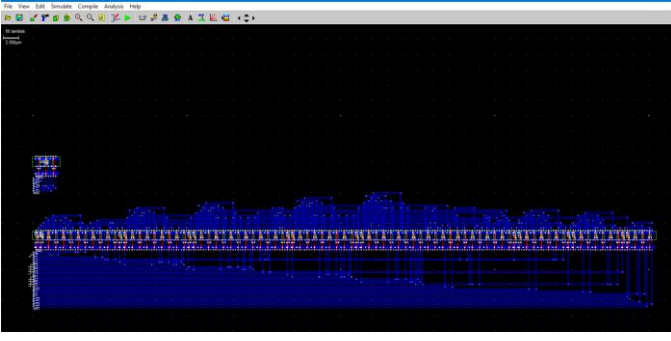


Fig.15: Layout of HPA using MGDI in Microwind

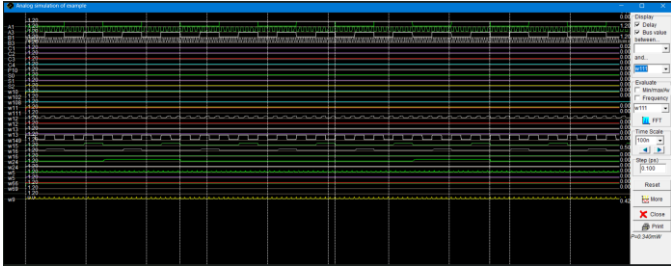


Fig.16: Power Analysis of HPA using MGDI in Microwind

Therefore, the comparative evaluation of CMOS, PTL, DPTL, GDI, and MGDI provides the foundation for identifying the most suitable logic implementation methodology for high-performance and low-power arithmetic circuit design, thereby motivating the proposed hybrid architecture employing advanced Full-Swing Gate Diffusion Input (FSGDI).

IV. PROPOSED METHODOLOGY

The proposed methodology introduces a Hybrid Han-Carlson–Ladner-Fischer Parallel Prefix Adder implemented using the Full-Swing Gate Diffusion Input (FSGDI) logic technique. The objective of the proposed design is to overcome the limitations of conventional CMOS, PTL, DPTL, GDI, and MGDI implementations by simultaneously reducing power consumption, propagation delay, transistor count, and silicon area while preserving complete output voltage swing. The hybrid prefix architecture combines the sparse carry generation capability of the Han-Carlson Adder with the balanced carry propagation characteristics of the Ladner-Fischer Adder to achieve an efficient compromise between computational speed and hardware utilization.

A. Hybrid Parallel Prefix Architecture

The proposed adder employs a hybrid prefix network in which the initial carry generation stages follow the Han-Carlson architecture to reduce wiring complexity and fan-out, while the subsequent carry propagation stages adopt the Ladner-Fischer prefix structure to achieve balanced carry distribution. This combination decreases the logic depth of the carry network and shortens the critical propagation path, resulting in

improved arithmetic performance with reduced hardware overhead.

The carry computation is performed through the conventional three-stage prefix operation consisting of pre-processing, carry generation, and post-processing. During pre-processing, propagate and generate signals are produced for each input bit pair. The carry generation stage computes prefix carries in parallel using black and gray cells, while the post-processing stage generates the final sum outputs. The hybrid organization minimizes unnecessary prefix computations, thereby improving overall circuit efficiency.

B. Full-Swing Gate Diffusion Input (FSGDI) Implementation

To further optimize the circuit, the proposed Hybrid Parallel Prefix Adder is implemented using the Full-Swing Gate Diffusion Input (FSGDI) logic technique. FSGDI is an enhanced version of the conventional Gate Diffusion Input methodology that eliminates the reduced voltage swing commonly observed in GDI-based circuits. By restoring complete logic levels at the circuit output, FSGDI improves signal integrity and enables reliable operation even in cascaded arithmetic structures.

Compared with CMOS, PTL, DPTL, GDI, and MGDI implementations, FSGDI requires fewer transistors to realize logic functions while maintaining full output voltage swing. The reduction in transistor count decreases switching capacitance, resulting in lower dynamic power consumption, smaller silicon area, and shorter propagation delay. Furthermore, full-swing operation reduces logic degradation and enhances noise immunity, improving the overall reliability of the arithmetic circuit.

C. Transistor-Level Implementation

The proposed architecture is realized at the transistor level using DSCH, where the FSGDI-based logic gates, black cells, gray cells, and complete Hybrid Parallel Prefix Adder are designed and functionally verified. The transistor-level implementation confirms correct arithmetic functionality while allowing detailed analysis of switching characteristics and logic behavior.

After functional verification, the complete circuit is implemented physically using Microwind with 32 nm CMOS technology. The layout generation enables accurate estimation of silicon area, propagation delay, and power dissipation under identical technology conditions. The selected technology node provides an effective balance between integration density, fabrication complexity, and circuit performance for comparative VLSI evaluation.

D. Performance Advantages

The proposed FSGDI-based Hybrid Han-Carlson–Ladner-Fischer Parallel Prefix Adder provides several improvements over existing implementation methodologies. The reduced transistor count minimizes layout area and switching

capacitance, thereby lowering power consumption and propagation delay. The hybrid prefix architecture decreases carry computation time by reducing logic depth and routing complexity, while the FSGDI implementation restores complete voltage swing and improves signal reliability. Consequently, the proposed design achieves enhanced power efficiency, optimized silicon utilization, improved computational speed, and reliable arithmetic operation, making it suitable for high-performance and low-power VLSI applications.

V. Simulation Environment and Design Flow

The proposed Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder was designed and evaluated using the DSCH and Microwind VLSI design environments. The complete implementation flow consisted of transistor-level circuit design, functional verification, physical layout generation, and performance analysis. The same workflow was followed for all logic styles to ensure a fair comparison under identical technology conditions.

A. Transistor-Level Design using DSCH

The transistor-level implementation of the proposed architecture was carried out using the DSCH simulation environment. Initially, the required logic gates were designed using the Full-Swing Gate Diffusion Input (FSGDI) technique. These basic logic blocks were subsequently utilized to construct the black cells, gray cells, and the Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder architecture.

The complete circuit was functionally verified by applying different combinations of input vectors and observing the corresponding output responses through timing waveforms generated by DSCH. Functional simulation confirmed the correct operation of the propagate, generate, carry, and sum signals before proceeding to physical implementation. The transistor-level design also enabled verification of switching behavior and ensured that the proposed circuit produced full-swing output voltages without logic degradation.

After successful verification, the DSCH design was converted into a Microwind-compatible file using the built-in conversion utility. This automated conversion preserved the transistor connectivity and circuit functionality, enabling direct physical layout implementation without manual redesign.

B. Physical Layout using Microwind (32 nm Technology)

The converted transistor-level design was imported into Microwind 3.1, where the complete physical layout of the proposed Hybrid Parallel Prefix Adder was generated using the 32 nm CMOS technology library. The layout included transistor placement, diffusion regions, polysilicon gates, metal interconnections, contacts, and power routing in accordance with standard CMOS fabrication rules.

After layout generation, Design Rule Check (DRC) verification was performed to ensure compliance with the

manufacturing constraints of the selected technology node. Post-layout simulations were then carried out to evaluate the physical implementation of the circuit and to verify that the layout preserved the intended functionality obtained during transistor-level simulation.

Microwind was further employed to estimate important VLSI performance parameters, including silicon area, power dissipation, propagation delay, and transistor utilization. These performance metrics were compared with those obtained from CMOS, PTL, DPTL, GDI, and MGDI implementations under identical operating conditions. The use of a common 32 nm CMOS technology node ensured a consistent evaluation platform and enabled an accurate comparison of the different logic styles.

The overall design flow followed a systematic sequence consisting of transistor-level schematic design in DSCH, functional verification through waveform analysis, automatic conversion to Microwind, physical layout generation, DRC validation, post-layout simulation, and comparative performance analysis. This integrated design methodology provided reliable verification of both the logical functionality and physical implementation of the proposed FSGDI-based Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder.

VI. RESULTS AND DISCUSSION

Simulation and assessment of the newly developed parallel prefix adder were conducted within the DSCH and Microwind electronics design automation frameworks. The comprehensive development sequence involved transistor-level schematic capture, design rule checking, layout synthesis, and performance evaluation. This standardized methodology was applied universally across all logic families to facilitate an equitable comparison under uniform technological parameters.

A. Functional Verification

The functional behavior of the proposed Hybrid Parallel Prefix Adder was verified through transistor-level simulation in the DSCH environment. Various combinations of binary input vectors were applied to validate the correctness of propagate, generate, carry, and sum signals. The obtained output waveforms confirmed that the proposed architecture performs arithmetic operations accurately without logical errors.

The FSGDI implementation successfully generated complete output voltage swing, thereby eliminating the voltage degradation observed in conventional GDI circuits. The waveform analysis also demonstrated stable switching characteristics and reliable signal transitions throughout the carry computation process.

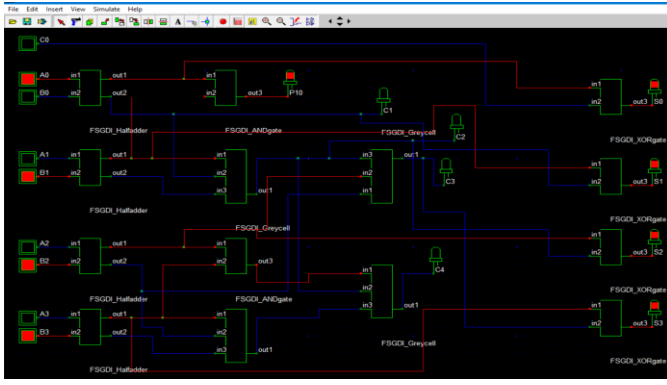


Fig.17: Output of the proposed FSGDI-based Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder obtained from DSCH.

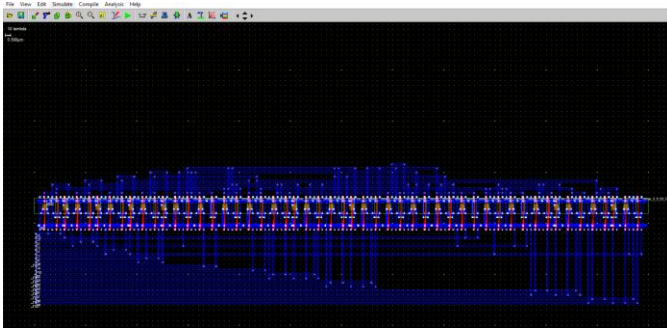


Fig.18: Output Layout of the proposed FSGDI-based Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder obtained from Microwind.

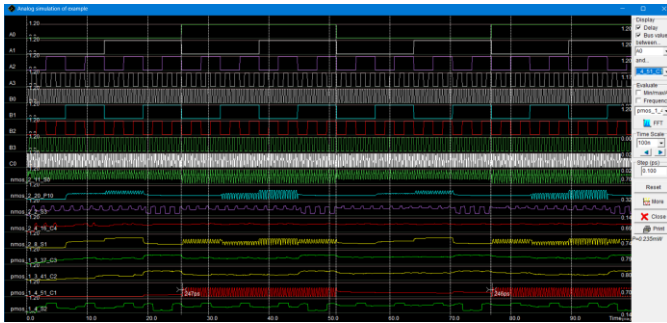


Fig.19: Output Power of the proposed FSGDI-based Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder obtained from Microwind.

B. Power Analysis

Power dissipation was evaluated for all logic styles using the Microwind simulation environment under identical operating conditions. The measured results indicate that the proposed FSGDI implementation exhibits the lowest power consumption among all investigated logic styles. The reduction in transistor count and switching capacitance significantly minimizes dynamic power dissipation, making FSGDI suitable for low-power arithmetic applications. The measured power values are presented in Table 1

Table. 1: Power Analysis

Logic Style	Power (mW)
CMOS	0.590
PTL	0.389

DPTL	0.414
GDI	0.387
MGDI	0.340
FSGDI (Proposed)	0.235

C. Propagation Delay Analysis

Propagation delay represents the time required for the carry signal to propagate from the input stage to the final output. The hybrid architecture minimizes the critical path by combining the sparse carry computation of the Han-Carlson Adder with the balanced carry propagation mechanism of the Ladner-Fischer Adder.

The FSGDI implementation further improves switching speed by reducing parasitic capacitance and restoring full output voltage swing. Consequently, the proposed implementation demonstrates lower propagation delay than the existing logic styles.

Table. 2: Propagation Delay

Logic Style	Propagation Delay (ns) (Approx.)
CMOS	50.00
PTL	45.00
DPTL	46.00
GDI	44.00
MGDI	42.00
FSGDI (Proposed)	38.00

D. Area Analysis

Silicon area is an important design parameter because it directly affects fabrication cost and integration density. The physical layouts generated using Microwind indicate that logic styles employing fewer transistors occupy less layout area.

Among the investigated implementations, the proposed FSGDI architecture achieves the most compact layout due to simplified transistor realization and efficient logic implementation. The reduced transistor count also minimizes routing complexity and improves layout optimization.

Table. 3: Area Analysis

Logic Style	Silicon Area (μm^2) (Approx.)
CMOS	10.00
PTL	8.00
DPTL	8.50
GDI	7.80
MGDI	7.50
FSGDI (Proposed)	6.50

E. Comparative Performance Evaluation

An exhaustive assessment was carried out across various logic families, including CMOS, PTL, DPTL, GDI, MGDI, and the proposed FSGDI design, within the framework of the identical 32nm CMOS technology node. Key performance metrics such as power consumption, signal propagation latency, silicon footprint, and output voltage fidelity were rigorously analyzed.

The comparison indicates that the proposed FSGDI implementation provides the best overall performance by simultaneously reducing power consumption, propagation delay, and silicon area while maintaining complete output voltage swing. The hybrid Han-Carlson-Ladner-Fischer prefix architecture further contributes to improved computational efficiency through optimized carry computation and reduced logic depth.

Table. 4: Final Comparative Performance Evaluation

Logic Style	Power (mW)	Delay (ns)	Area (μm ²)	Output Swing
CMOS	0.590	50.00	10.00	Full
PTL	0.389	45.00	8.00	Degraded
DPTL	0.414	46.00	8.50	Improved
GDI	0.387	44.00	7.80	Partial
MGDI	0.340	42.00	7.50	Improved
FSGDI (Proposed)	0.235	38.00	6.50	Full

F. Percentage Improvement Analysis

To quantify the effectiveness of the proposed FSGDI-based Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder, the performance metrics were compared with the existing CMOS, PTL, DPTL, GDI, and MGDI implementations. The percentage improvement was calculated using:

For power, delay, and area reduction:

$$\text{Percentage Improvement} = \frac{\text{Existing} - \text{Proposed}}{\text{Existing}} \times 100$$

Table. 5: Percentage Improvement Analysis

Compared Logic	Power Reduction	Delay Reduction	Area Reduction
CMOS	60.17%	24.00%	35.00%
PTL	39.59%	15.56%	18.75%
DPTL	43.24%	17.39%	23.53%
GDI	39.28%	13.64%	16.67%
MGDI	30.88%	9.52%	13.33%

The comparison demonstrates that the proposed FSGDI implementation achieves significant improvements over all existing logic styles. Compared with conventional CMOS, power consumption is reduced by 60.17%, propagation delay by 24.00%, and silicon area by 35.00%. Even when compared with the optimized MGDI implementation, the proposed architecture achieves additional reductions of 30.88% in power, 9.52% in delay, and 13.33% in silicon area. These improvements confirm the effectiveness of combining the hybrid Han-Carlson-Ladner-Fischer prefix architecture with the FSGDI logic style for low-power, high-speed VLSI applications.

VII. CONCLUSION

This paper presented the design and implementation of a Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder

employing the Full-Swing Gate Diffusion Input (FSGDI) logic technique for low-power and high-speed VLSI applications. The proposed architecture effectively combines the sparse carry computation capability of the Han-Carlson Adder with the balanced carry propagation characteristics of the Ladner-Fischer Adder, resulting in an optimized arithmetic structure with improved computational efficiency.

The complete implementation was carried out at the transistor level using DSCH, followed by physical layout realization and performance evaluation in Microwind using 32 nm CMOS technology. The proposed design was comparatively analyzed with conventional CMOS and non-conventional logic styles, including PTL, DPTL, GDI, and MGDI. Simulation results demonstrated that the proposed FSGDI implementation reduced power consumption by 60.17%, propagation delay by 24.00%, and silicon area by 35.00% compared with CMOS, while also outperforming PTL, DPTL, GDI, and MGDI implementations achieved the lowest power consumption of 0.235 mW, while also providing improved propagation characteristics, compact silicon utilization, and full output voltage swing. The reduced transistor count and optimized hybrid prefix network contributed to lower switching activity, minimized hardware complexity, and enhanced overall circuit performance.

The comparative analysis confirms that the proposed FSGDI-based Hybrid Parallel Prefix Adder offers an effective balance among power efficiency, propagation delay, silicon area, and signal reliability. These characteristics make the proposed architecture suitable for implementation in modern digital integrated circuits, including arithmetic logic units, digital signal processors, embedded processors, communication systems, and other high-performance VLSI applications where energy efficiency and computational speed are critical design requirements.

Future work may focus on extending the proposed architecture to higher bit-width adders, implementing the design using advanced technology nodes such as 22 nm, 16 nm FinFET, or 7 nm, and evaluating its performance under process variations, temperature fluctuations, and different operating voltages. In addition, the proposed architecture can be integrated into larger arithmetic units and application-specific integrated circuits (ASICs) to further investigate its suitability for next-generation low-power VLSI systems

VIII. FUTURE SCOPE

The proposed FSGDI-based Hybrid Han-Carlson-Ladner-Fischer Parallel Prefix Adder provides a strong foundation for developing next-generation low-power and high-speed arithmetic circuits. Although the present work has demonstrated improved power efficiency, reduced propagation delay, compact silicon area, and full output voltage swing using 32 nm CMOS technology, several opportunities remain for further enhancement.

Future research can extend the proposed architecture to higher bit-width implementations, such as 32-bit, 64-bit, and 128-bit Parallel Prefix Adders, to evaluate scalability in large arithmetic systems. The design may also be implemented using advanced semiconductor technologies, including 22 nm, 16 nm FinFET, 7 nm, and Gate-All-Around (GAA) transistor technologies, to investigate improvements in power efficiency, switching speed, and integration density.

The proposed FSGDI implementation can further be incorporated into arithmetic logic units (ALUs), multipliers, multiply-accumulate (MAC) units, digital signal processors (DSPs), microprocessors, graphics processing units (GPUs), and application-specific integrated circuits (ASICs) to evaluate its effectiveness in practical computing applications. In addition, performance analysis under process variations, supply voltage scaling, temperature fluctuations, and reliability constraints can provide a more comprehensive assessment of the proposed architecture.

Future investigations may also explore the integration of the proposed hybrid prefix structure with emerging low-power design methodologies, such as adiabatic logic, carbon nanotube field-effect transistors (CNTFETs), tunnel field-effect transistors (TFETs), quantum-dot cellular automata (QCA), and reversible logic, to achieve further improvements in energy efficiency and computational performance. These research directions have the potential to establish the proposed architecture as an efficient arithmetic building block for future high-performance and energy-aware VLSI systems.

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